

Question 1:

Solve the following problem with CPU scheduling algorithms.

Draw a table with 4 rows & 3 columns having the following details.

Process	AT	BT
P ₁	0	5
P ₂	0	4
P ₃	0	3

There is a Context Switch in the CPU
the Context Switch time is 5 seconds.
Consider the Context Switch time below
Processes of five seconds.

Apply the following Algorithms:-

- (i) preemptive algorithms:
 - (i) SRTF
 - (ii) Priority scheduling

2) Non-preemptive Scheduling Algorithm (FCFS)

(i) FCFS

(ii) SJF

(iii) Priority Scheduling

Solve each algorithm with its Gantt chart, CT, TAT, WT, average TAT and average WT

Given,

Process

AT

BT

P₁

0

5

P₂

0

4

P₃

0

3

→ Context Switch time = 5 sec

$$TAT = CT - AT$$

$$WT = TAT - BT$$

(i) FCFS

Order: P₁ → P₂ → P₃

Process	AT	BT	CT	TAT	WT
P ₁	0	5	22	22	17
P ₂	0	4	12	12	8
P ₃	0	3	3	3	0

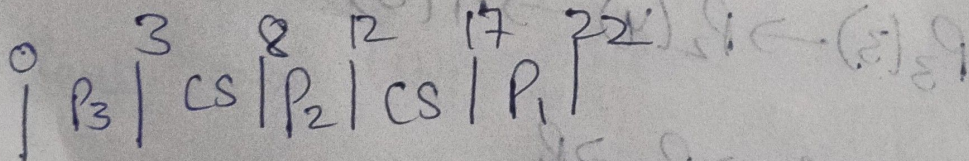
$$\text{Average WT} = \frac{17+8+0}{3} = 8.33$$

$$\text{Average TAT} = \frac{22+12+3}{3} = 12.33$$

(iii) SRTF

Order: P₃ → P₂ → P₁

Gantt chart:



Process	AT	BT	CT	TAT	WT
P ₁	0	5	22	22	17
P ₂	0	4	12	12	8
P ₃	0	3	3	3	0

Average TAT = 12.33

(iv) priority scheduling:

Cannot solve because priority values are not given.

question 2:

A mobile operating system has the following tasks

Task	AT	CPU Burst
Music Player (MP)	0	20
Touch Event (TE)	3	<u>1</u>
whatsapp Notification (WN)	5	2
Camera Launch (CL)	6	4

Questions:

(i) which scheduling algorithm is most suitable
FCFS, SJF, SRTF (or) RR?

(ii) Draw the execution timeline for
SRTF

(iii) Explain why the selected algorithm
improves user experience

(iv) which task receives the fastest response
time

(i) SRTF is the suitable scheduling
algorithm because SRTF always selects
the task with shortest remaining CPU
burst time

(ii) SRTF Execution Timeline

In SRTF, the process with the shortest
remaining burst time is selected.

2-3: MP runs

Remaining MP = 20 - 3 = 17

At 3: TE arrives with BT = 1

TE is shorter than MP $\rightarrow$ TE runs from 3-4

4-5: MP continues

Remaining MP = 16

At 5: WN arrives with BT = 2

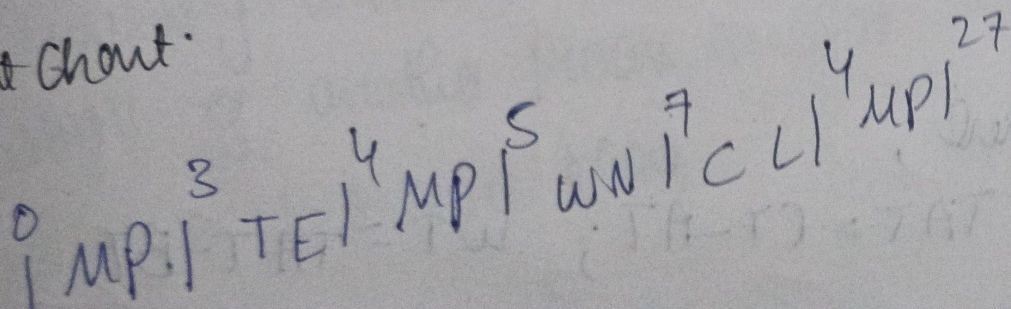
WN is shorter than MP $\rightarrow$ WN runs

5-7: - WN completes

7-11: Compare MP = 16 & CL = 4 $\rightarrow$ CL runs

11-27: CL finishes, so remaining MP runs until completion.

Grant Chart:



→ Final Execution Order

MP → TE → MP → WN → CL → MP

(iii) SRTF gives quick response to short tasks such as touch events, notifications and camera launch.

∴ therefore, the mobile device feels:-

- * Fast
- * ~~Responsive~~ Responsive
- * Interactive

(iv) Response Time = First CPU start time - AT

TASK	AT	BT	first CPU start	CT	TAT	WT	RT
			0	27	27	7	0
MP	0	20	3	4	1	0	0
TE	3	1	5	7	2	0	0
WN	5	2	7	11	5	1	1
CL	6	4					

where,

$$TAT = CT - AT; \quad WT = TAT - BT$$

Question 3:
Define the following terms:-

(i) Starvation
Starvation occurs when a process waits for a very long time because other processes are repeatedly selected for execution.

(ii) Convoy effect
Convoy effect occurs in FCFS scheduling when a long process makes many short processes wait behind it.

(iii) Context Switching.
It is the process of saving the state of the currently running process and loading the state of another process so that the CPU can switch between them.

(iv) Aging

Aging is a technique used to prevent starvation by gradually increasing the priority of the process that has been waiting for a long time.

Question 4:

Consider the following process-

Process	AT	BT	Priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	8 9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

(i) FCFS

FCFS

means

first come first serve

$\begin{array}{c} 0 \quad 4 \quad 13 \quad 19 \quad 23 \quad 32 \quad 38 \quad 42 \quad 51 \\ | P_1 | P_2 | P_3 | P_4 | P_5 | P_6 | P_7 | P_8 | P_9 \end{array}$

Process	AT	BT	CT	TAT	WT	RT
P_1	0	4	4	4	0	0
P_2	0	9	13	13	4	4
P_3	0	6	19	19	13	13
P_4	0	4	23	23	19	19
P_5	0	9	32	32	23	23
P_6	0	6	38	38	32	32
P_7	1	4	42	41	37	37
P_8	1	9	51	50	41	41
P_9	2	6	57	55	49	49

Average TAT = 30.56

Average WT = 24.22

(ii) SJF means Shortest Job first. Since it is non-preemptive, once a process starts, it finishes completely.

Order:

Shortest available processes are selected

$P_1 \rightarrow P_4 \rightarrow P_7 \rightarrow P_3 \rightarrow P_6 \rightarrow P_9 \rightarrow P_2 \rightarrow P_5 \rightarrow P_8$

Gantt chart

0 4 8 12 18 24 30 39 48 57
 $| P_1 | P_4 | P_7 | P_3 | P_6 | P_9 | P_2 | P_5 | P_8 |$

Process	AT	BT	CT	TAT	WT	RT
P_1	0	4	4	4	0	0
P_4	0	4	8	8	4	4
P_7	1	4	12	11	7	7
P_3	0	6	18	18	12	12
P_6	0	6	24	24	18	18
P_9	2	6	30	28	22	22
P_2	2	9	39	39	30	30
P_5	0	9	48	48	39	39
P_8	1	9	56	56	47	47

Average
Average WT = 19.89

(iii) SRTF

SRTF means shortest Remaining Time first

→ It is the preemptive Version of SJF

Process	CT	TAT	WT	RT
	4	4	0	0
P ₁			4	4
P ₄	8	8	7	7
P ₇	12	11	12	12
P ₃	18	18	18	18
P ₆	24	24	22	22
P ₉	30	28	30	30
P ₂	39	39	39	39
P ₅	48	48	47	47
P ₈	57	56		

Average TAT = 26.22

Avg WT = 19.89